

Physics - 2105

3rd class - final

29-03-2023

Coulombs law:

$$F = k \frac{q_1 q_2}{r^2}$$

q_1, q_2 → magnitude of charge.

r → distance between charge.

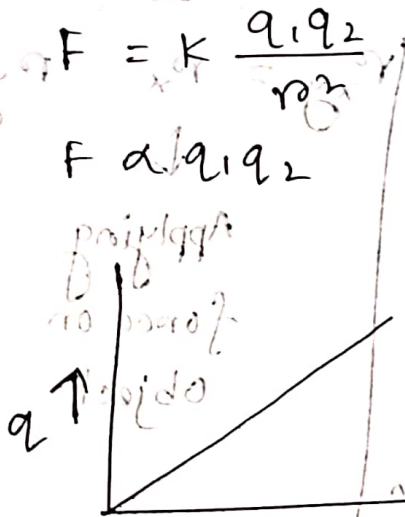
must be read Coulombs statement

$$K = C = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$$

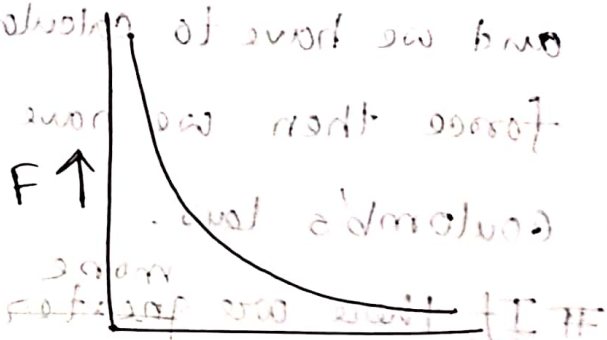
$$K = \frac{1}{4\pi\epsilon_0}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

Draw graph for Coulombs law:



$$F \propto \frac{1}{r^2}$$



Another name of Coulomb's law is Coulomb's inverse law.

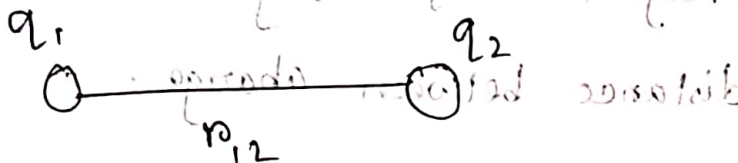
Coulomb's Constant Calculation.

11 law (vector)

Physics- 2105

4th class - final

02/04/2023



Force Applied
on q_1

$$\vec{F}_{12} = \frac{k q_1 q_2}{r_{12}^2} \times \hat{r}_{12}$$

Force Applied
on q_2

$$\vec{F}_{21} = \frac{k q_1 q_2}{r_{21}^2} \times \hat{r}_{21}$$

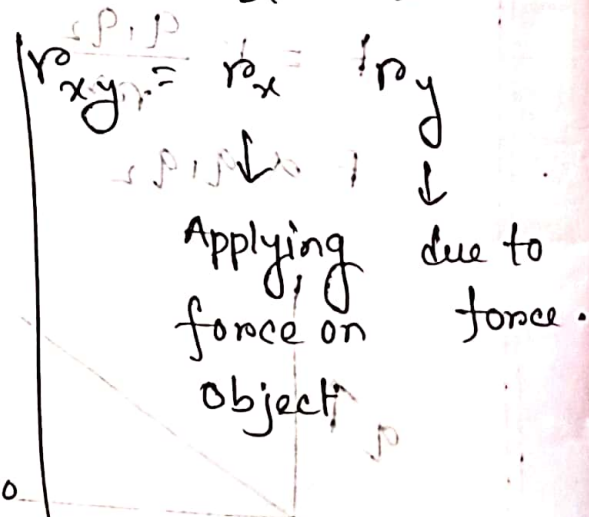
$$\hat{r}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$$

$$\vec{r}_{12} = \vec{r}_1 - \vec{r}_2$$

$$\vec{r}_{21} = \vec{r}_2 - \vec{r}_1$$

If there are two charge and we have to calculate the force then we have to use Coulomb's law.

If there are ^{more} greater than two charge and we have to calculate the force then we have to use superposition principle.



Superposition

Application of Coulomb's law:

1. ~~at~~ To calculate electrical force.
2. " " Coulomb's Constant (K/c)
3. " " distance between atom.
4. " " electrical Potential Energy.
5. " " field ($\vec{E} = \frac{\vec{F}}{q}$).
6. " " vacuum permittivity (ϵ_0)
7. " " Gauss's law.
8. Used in Quantum field theory origin.

Quantum means 10^{-5}

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ବି. ମାଧ୍ୟମରେ ଥାଏ.

$$\left\{ (0.01 \times 0.1) \times (5.8 \times 10^{-8}) \times 0.5 \right\}$$

$$= 0.01 \times 10^{-1}$$

Superposition

Application of Coulomb's law:

1. ~~at~~ To calculate electrical force.
2. " " Coulomb's Constant (K/C)
3. " " distance between atom.
4. " " electrical Potential Energy.
5. " " field ($\vec{E} = \frac{\vec{F}}{q}$).
6. " " vacuum permittivity (ϵ_0)
7. " " Gauss's law.
8. Used in Quantum field theory origin.

Quantum means $10^{-15}m$

କ୍ଷୁଦ୍ର କଣିକାମାନଙ୍କର ଗତିକୁ କ୍ୱାଣ୍ଟମ୍ କୁହାଯାଏ ।

କ୍ୱାଣ୍ଟମ୍ ମାଧ୍ୟମରେ ଗଣନା କରାଯାଏ ।

Example: 3 (1) What is the force between two 3gm pennies one meter apart if we remove all the electrons from the copper atoms?

(4) What is their acceleration as they separate?

$\Rightarrow$ (1) Total number of atoms

1 mole $\rightarrow 6 \times 10^{23}$

$\therefore 63.5 \text{ gm} \rightarrow 6 \times 10^{23} \text{ atom}$

$\therefore 1 \text{ gm} \rightarrow \frac{6 \times 10^{23}}{63.5} \text{ atom}$

$\therefore 3 \text{ gm} \rightarrow \frac{6 \times 10^{23} \times 3}{63.5} \text{ atom}$

$$= 2.83 \times 10^{22} \text{ atom}$$

$$\therefore q = ne$$

$$= zne$$

$$= \{ 29 \times (2.83 \times 10^{22}) \times (1.6 \times 10^{-19}) \} \text{ C}$$

$$= 1.31 \times 10^5 \text{ C}$$

$$F = \frac{k q_1 q_2}{r^2}$$

$$= \frac{k (q)^2}{r^2}$$

$$= \frac{(9 \times 10^9) \times (1.31 \times 10^{-5})^2}{1^2}$$

$$= 1.54 \times 10^{20} \text{ N}$$

$$(ii) F = ma$$

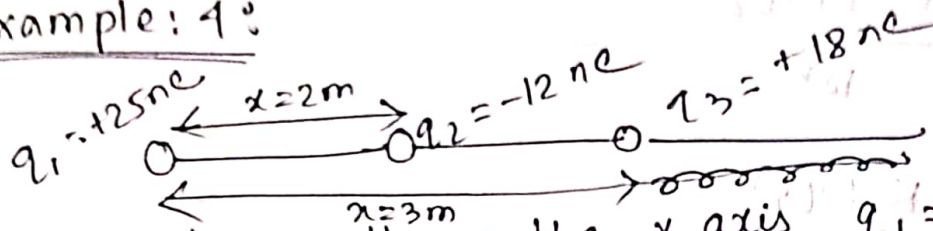
$$a = \frac{F}{m}$$

$$= \frac{1.54 \times 10^{20}}{3 \times 10^{-3}}$$

$$= 5.15 \times 10^{22} \text{ ms}^{-2}$$

Superposition Statement:

Example: 4:



Three charges lie on the x axis $q_1 = +25 \text{ nC}$ at the origin, $q_2 = -12 \text{ nC}$ at $x = 2 \text{ m}$, $q_3 = +18 \text{ nC}$ at $x = 3 \text{ m}$

What is the net force on q_1 ?

Force = attractive = -
(+ $\longleftrightarrow$ - = -)

Force = repulsive = +

(- $\longleftrightarrow$ - = +)
(+ $\longleftrightarrow$ + = +)

Example: 6: In an atom can we neglect the gravitational force between the electrons and protons?
What is the ratio of Coulomb's electric force to Newton's gravity force for 2 electrons separated by a distance r !

$$e^- = 1.6 \times 10^{-19} \text{ e.}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

Example : 13 :

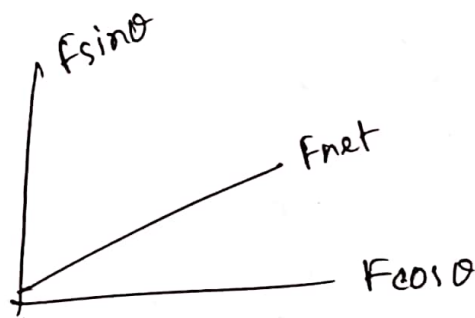
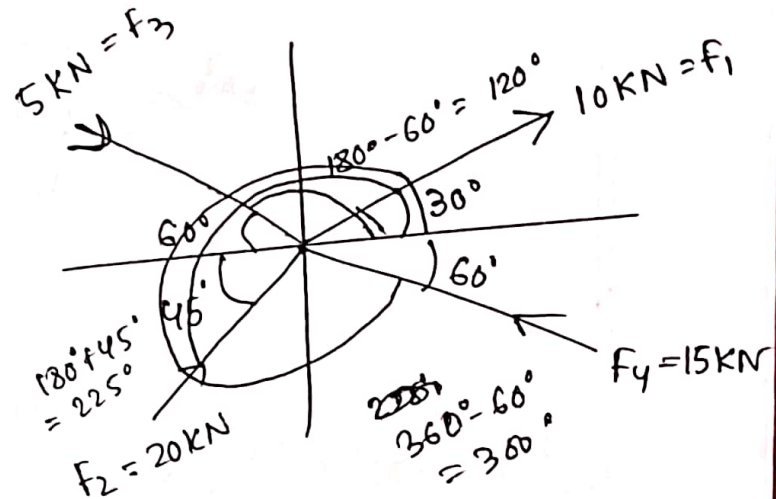
(71) Determine analytically the magnitude and direction of the resultant of the following four forces acting at a point.

(i) 10 kN pull N 30° E.

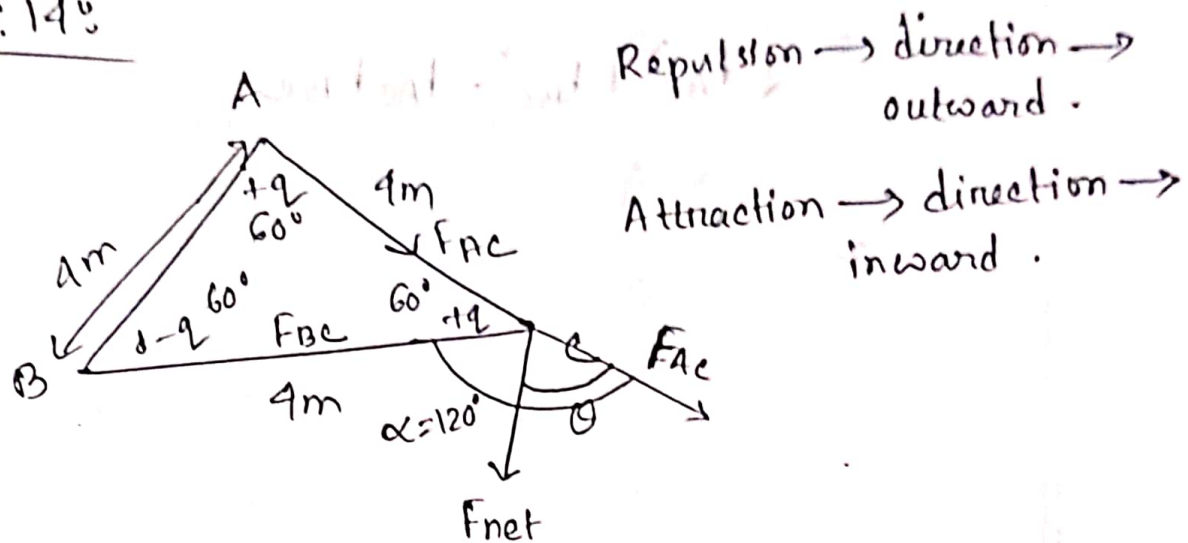
(ii) 20 kN push S 45° W.

(iii) 5 kN push N 60° W.

(iv) 15 kN push S 60° E.



Example: 14%



$$F_{AC} = k \frac{q_1 q_2}{r^2} = k \frac{+q \times +q}{r^2}$$

$$= (9 \times 10^9) \times \frac{(100 \times 10^{-6}) \times (100 \times 10^{-6})}{16}$$

$$F_{BC} = k \frac{+q \times -q}{r^2}$$

$$= k \frac{(100 \times 10^{-6}) \times -(100 \times 10^{-6})}{16}$$

$$F_{net} = R$$

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \alpha}$$

$$F_{net} = \sqrt{F_{AC}^2 + F_{BC}^2 + 2F_{AC} \cdot F_{BC} \cos 120^\circ}$$

$$= \sqrt{F_{AC}^2 + F_{BC}^2 + 2F_{AC} \cdot F_{BC} \cdot -\frac{1}{2}}$$

Resultant = $\sqrt{F_{Ac}^2 + F_{Bc}^2 - F_{Ac} F_{Bc} \cos 120^\circ}$



$$\therefore \theta = \tan^{-1} \left(\frac{Q \sin \alpha}{P + Q \cos \alpha} \right)$$

$$= \tan^{-1} \frac{5.625 \sin 120^\circ}{5.625 + 5.625 \cos 120^\circ}$$

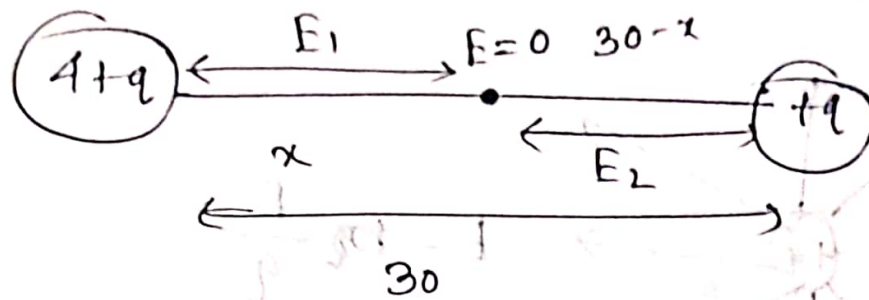
$$= \tan^{-1} \frac{5.625 \times \frac{\sqrt{3}}{2}}{5.625 \times \left(1 - \frac{1}{2} \right)}$$

$$= \tan^{-1} \frac{\sqrt{3} \times 5.625}{2} \times \frac{2}{5.625}$$

$$= \tan^{-1} (\sqrt{3})$$

$$= 60^\circ$$

Example: 8:



$$E_1 = E_2$$

$$\frac{kq_1}{r^2} = \frac{kq_2}{r^2}$$

$$\text{or, } \frac{k+4q}{x^2} = \frac{k+q}{(30-x)^2}$$

$$\text{or, } \frac{4}{x^2} = \frac{1}{(30-x)^2}$$

$$\text{or, } 4(30-x)^2 = x^2$$

$$\text{or, } 4(30^2 - 2 \cdot 30 \cdot x + x^2) = x^2$$

$$\text{or, } 4(900 - 60x + x^2) = x^2$$

$$\text{or, } 3600 - 240x + 4x^2 = x^2$$

$$\text{or, } 2(30-x) = x$$

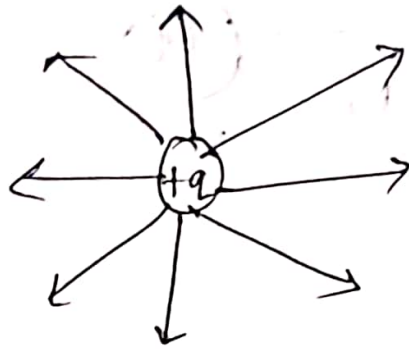
$$\text{or, } x = 60 - 2x$$

$$\text{or, } 2x + x = 60$$

$$\text{or, } 3x = 60$$

$$x = 20 \text{ cm}$$

Electrical Fields



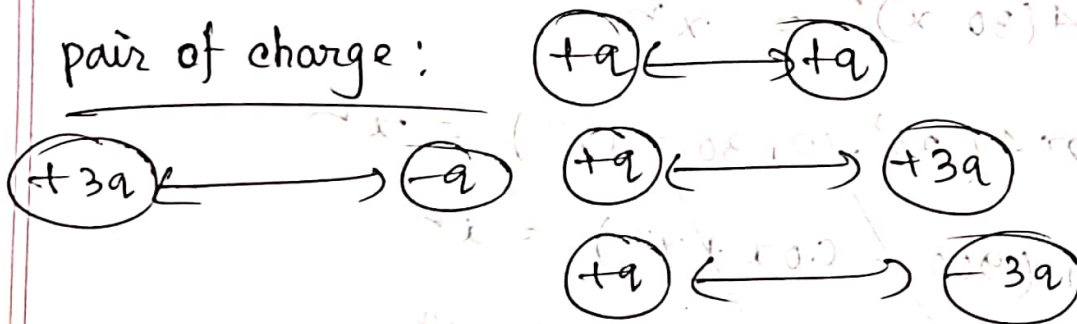
$$E = \frac{F}{q}$$

+ve charge $\rightarrow$ Field lines $\rightarrow$ Outward.

-ve " $\rightarrow$ Field lines $\rightarrow$ inward.

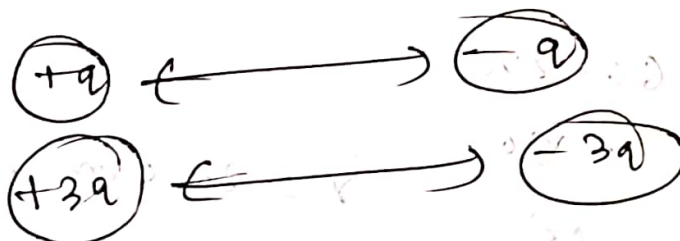


pair of charge:



Dipole charge: same magnitude.

Charge character opposite.



$$F = \frac{k q q_0}{r^2}$$

$$E = \frac{F}{q}$$

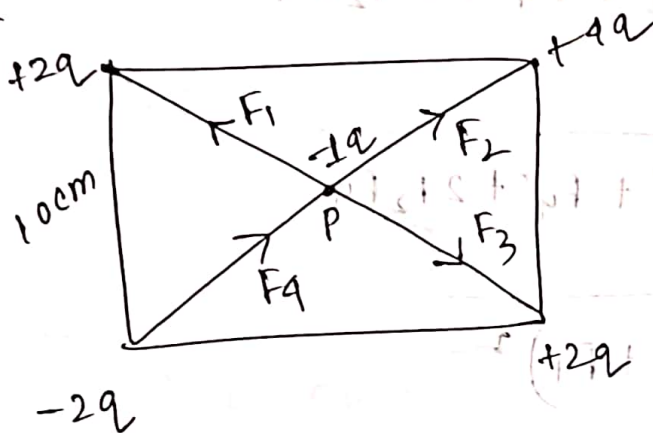
$$= \frac{k q q_0}{r^2} \times \frac{1}{q}$$

$$= \frac{k q_0}{r^2}$$

$$E = \frac{k q_0}{r^2}$$

Assignment: 3

Problem: 8:



$$q = 3 \times 10^{-9} \text{ C}$$

$$(1) F_{\text{net}} = ??$$

$$(11) \theta = ??$$

for square diagonal:

$$d = a\sqrt{2}$$

$$r = \frac{d}{2} = \frac{a\sqrt{2}}{2} = 0.0707$$

$$F_{13} = F_3 - F_1$$

$$= \frac{k \cdot 2q \times -1q}{r^2} - \frac{k \cdot 2q \times -1q}{r^2}$$

$$= 0.$$

$$\vec{F}_{24} = \vec{F}_2 + \vec{F}_4, \alpha = 0^\circ$$

$$|\vec{F}_{24}| = \sqrt{F_2^2 + F_4^2 + 2 \cdot F_2 F_4 \cos \alpha}$$

$$= \sqrt{F_2^2 + F_4^2 + 2 F_2 F_4}$$

$$= \sqrt{(F_2 + F_4)^2}$$

$$= F_2 + F_4$$

$$= k \frac{4q \times q}{r^2} + \frac{k \cdot 2q \times q}{r^2}$$

$$= \frac{kq}{r^2} (4q + 2q)$$

$$= \frac{kq}{r^2} 6q = \frac{k \cdot 6q^2}{r^2}$$

$$= \frac{(9 \times 10^{-9})}{(0.0707)^2} \times 6 \times (3 \times 10^{-9})$$

$$= 32409.78776$$

$$F_{net} = \sqrt{F_{13}^2 + F_{24}^2}$$

$$= \sqrt{0^2 + F_{24}^2}$$

$$= F_{24}$$

$$= 32409.78776$$

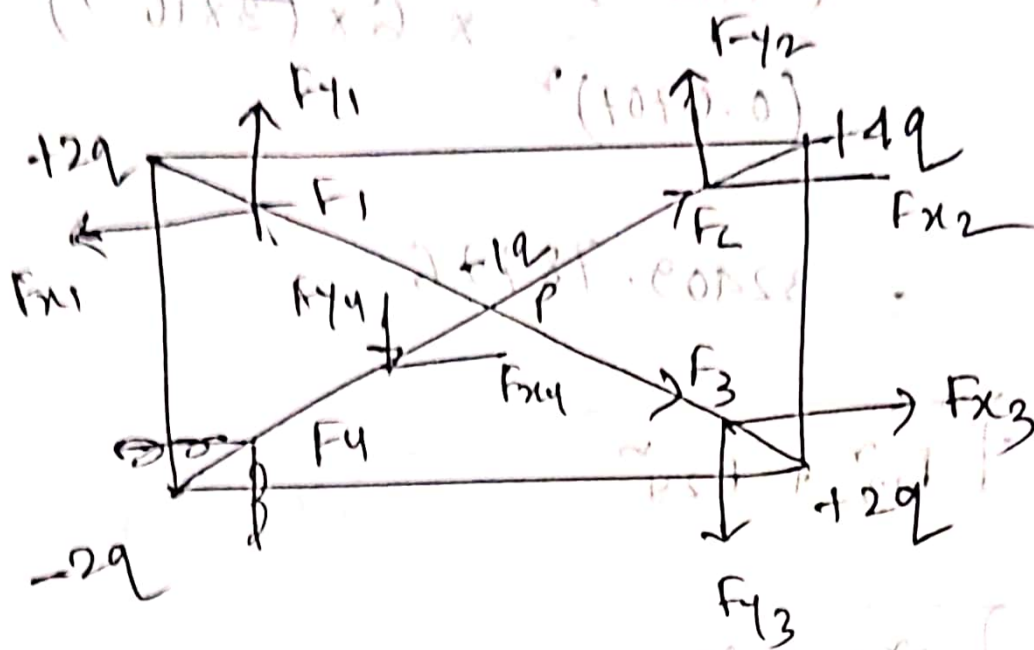
$$\theta = \tan^{-1} \left(\frac{Q \sin \alpha}{P + Q \cos \alpha} \right)$$

$$= \tan^{-1} \frac{F_{24} \sin 0^\circ}{F_{13} + F_{24} \cos 0^\circ}$$

$$= \tan^{-1} 0$$

$$= 0^\circ$$

Alternative:



$$F_x = -F_{x1} \cos \theta + F_{x2} \cos \theta + F_{x3} \cos \theta + F_{x4} \cos \theta$$

$$F_y = F_{y1} \sin \theta + F_{y2} \sin \theta - F_{y3} \sin \theta + F_{y4} \sin \theta$$

$$F_x = -k \frac{+2q \times q}{r^2} + \frac{k \frac{4q \times q}{r^2} \cos \theta}{\cancel{\cos \theta}} + \frac{k \frac{2q \times q}{r^2}}{\cancel{\cos \theta}} + \frac{k \frac{2q \times q}{r^2} \cos \theta}{\cancel{\cos \theta}}$$

$$= \frac{kq}{r^2} (4q + 2q)$$

$$= \frac{kq}{r^2} 6q$$

$$= \frac{kq^2}{r^2} 6$$

F_y : solve it yourself.

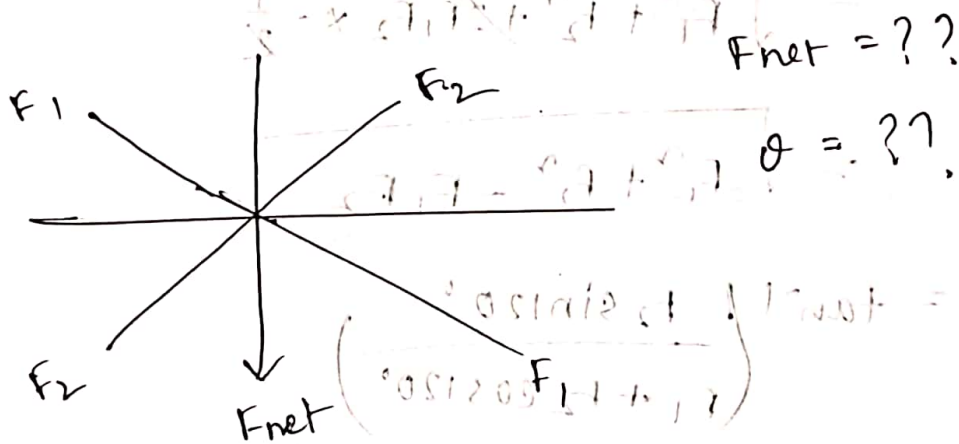
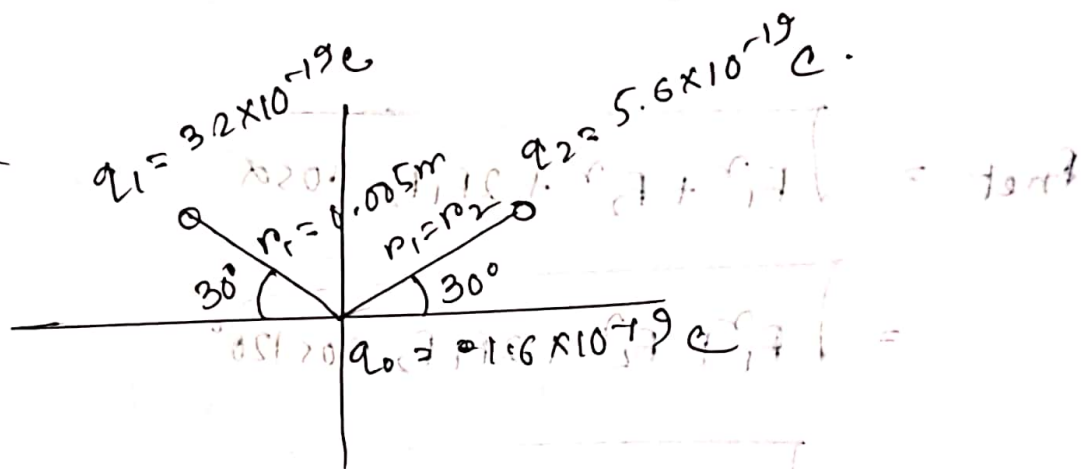
$$F_{net} = \sqrt{F_x^2 + F_y^2} = "$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$= \tan^{-1} \left(\frac{F_y}{F_x} \right)$$

$$= \tan^{-1} \left(\frac{(0.01 \times 2.1) \times (0.01 \times 2.2)}{(200.0)} \right)$$

Problem 10:



$$F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$= \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \theta}$$

$$F_1 = \frac{k (3.2 \times 10^{-19}) \times (1.6 \times 10^{-19})}{(0.005)^2}$$

$$F_2 = \frac{k (5.6 \times 10^{-19}) \times (1.6 \times 10^{-19})}{(0.005)^2}$$

$$F_{net} = \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos \alpha}$$

$$= \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \cos 120^\circ}$$

$$= \sqrt{F_1^2 + F_2^2 + 2F_1F_2 \times -\frac{1}{2}}$$

$$= \sqrt{F_1^2 + F_2^2 - F_1F_2}$$

$$\theta = \tan^{-1} \left(\frac{F_2 \sin 120^\circ}{F_1 + F_2 \cos 120^\circ} \right)$$

$$= \tan^{-1} \left(\frac{F_2 \times \frac{\sqrt{3}}{2}}{F_1 - \frac{1}{2}F_2} \right)$$

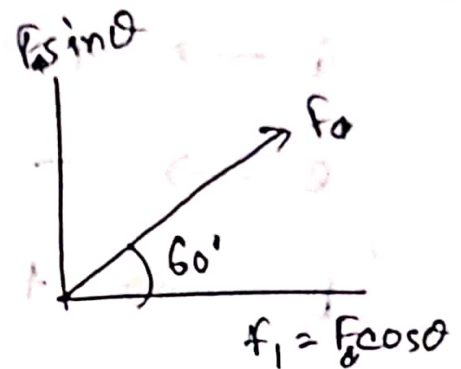
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Example 3: $\vec{F}_1 = 3\hat{i} - 4\hat{j}$

$$|\vec{F}_1| = \sqrt{3^2 + (-4)^2}$$

$$= \sqrt{9 + 16}$$

$$F_1 = 5$$



$$F_x = F \cos \theta = 5 \cos 60^\circ$$

$$F_y = F \sin \theta = 5 \sin 60^\circ$$

$$40 = F \cos 60^\circ$$

$$F = \frac{40}{\cos 60^\circ} = 80 \text{ N}$$

$$\therefore F_y = \text{vertical component} = 80 \times \sin 60^\circ = ??$$

$$\vec{F}_1 \cdot \vec{F}_2$$

$$= F_1 \cdot F_2 \cos \theta$$

$$\cos \theta = \frac{\vec{F}_1 \cdot \vec{F}_2}{F_1 F_2}$$

$$= \frac{(3\hat{i} - 4\hat{j}) \cdot (-2\hat{i} + 3\hat{k})}{5 \times \sqrt{13}} = -\frac{6}{5\sqrt{13}} \therefore \theta = \cos^{-1} \left(\frac{-6}{5\sqrt{13}} \right)$$

2

Direction of electrical parameter

$i \rightarrow +$ to $-$

$e^- \rightarrow -$ to $+$

$E \rightarrow +$ to $-$

$v \rightarrow$ no direction.

$F \rightarrow$ (1) Attractive $\rightarrow$ outward

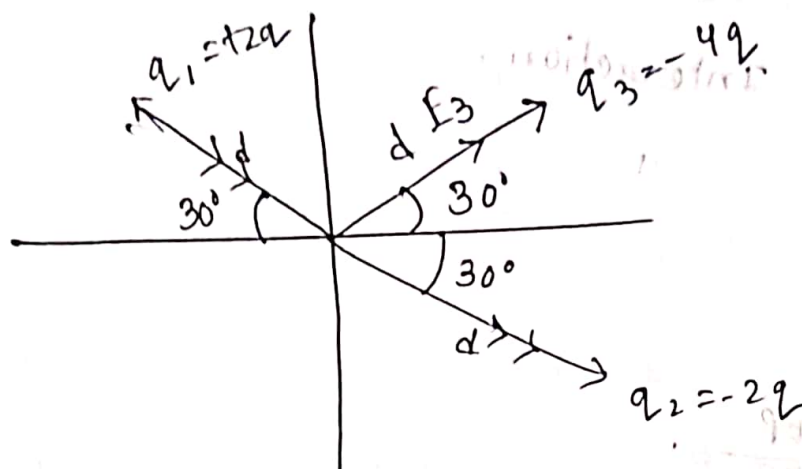
(-) Repulsive $\rightarrow$ inward

$p \rightarrow -$ to $+$

$+$ $\rightarrow$ outward

$-$ $\rightarrow$ inward

Problem : 1 : Find the resultant of the following electric field vectors.



$$E = \frac{F}{q} = k \frac{q}{r^2}$$

$$E_1 = k \frac{2q}{d^2}$$

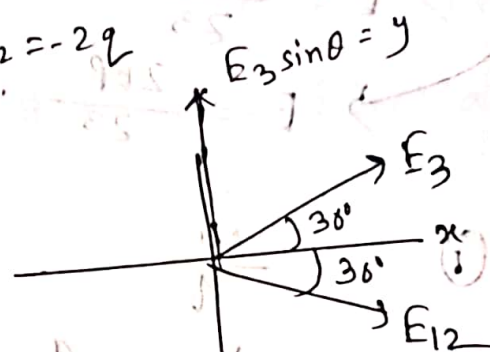
$$E_2 = k \frac{2q}{d^2}$$

$$E = E_1 + E_2 = k \frac{2q}{d^2} + k \frac{2q}{d^2}$$

$$= k \frac{4q}{d^2}$$

$$E_3 = k \frac{4q}{d^2}$$

$$E_3 = E_{12}$$



$$E_{3x} = E_3 \cos \theta$$

$$E_{3y} = E_3 \sin \theta$$

$$E_{12x} = E_{12} \cos \theta$$

$$E_{12y} = E_{12} \sin \theta$$

$$E_x = E_{3x} + E_{12x} = E_3 \sin \theta$$

$$= E_3 \cos \theta + E_{12} \cos \theta$$

$$= E_3 \cos \theta + E_3 \cos \theta$$

$$= 2E_3 \cos \theta = 2E_{3x} \cos \theta$$

Electrical field due to Electric dipole

(i) horizontal Interaction,

(ii) vertical

$$E = \frac{KP}{23}$$

$$E = \frac{2KP}{23}$$

(i) $V = \frac{W}{q}$

(ii) $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$

(iii) $V = Ed = -Eq$

(iv) $V = \frac{u}{q} = \frac{\Delta u}{q}$

V = Electrical Potential

d = displacement

$$u = \dots$$

$$W = \dots$$

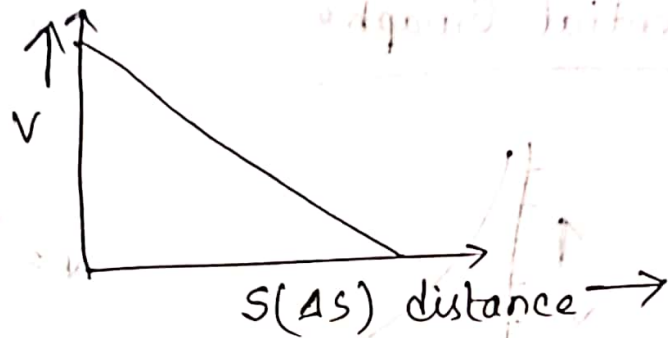
$$W = Vq = -Edq = -qEd = -q_0Ed$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$= (1.6 \times 10^{-19}) \times 1$$

$$= 1.6 \times 10^{-19} \text{ J}$$

$$E = - \frac{\Delta V}{\Delta s}$$



When a particle move from one place to another inside the Electric field that called Potential Energy.

Electric Potential Energy:

$$(1) V = \frac{u}{q} \quad \boxed{u = qV}$$

$$(2) \Delta u = -W = u_f - u_i$$

$$(3) u = K \frac{q_1 q_2}{r}$$



Potential Graph:

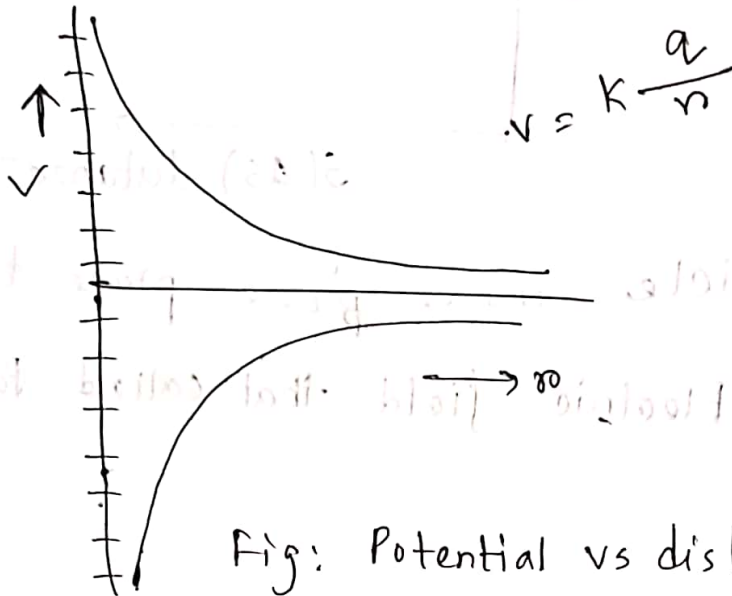
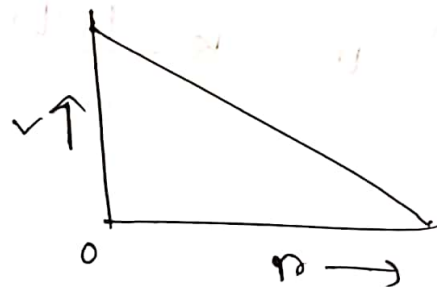
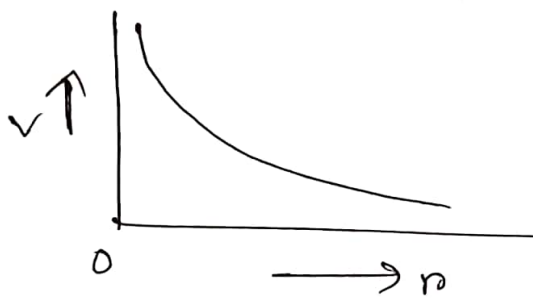


Fig: Potential vs distance -

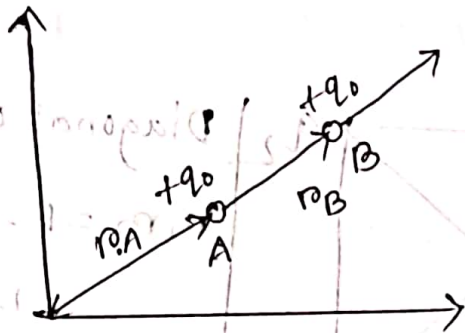
on
 V vs. S graph

on
 V vs r graph



Potential energy graph:

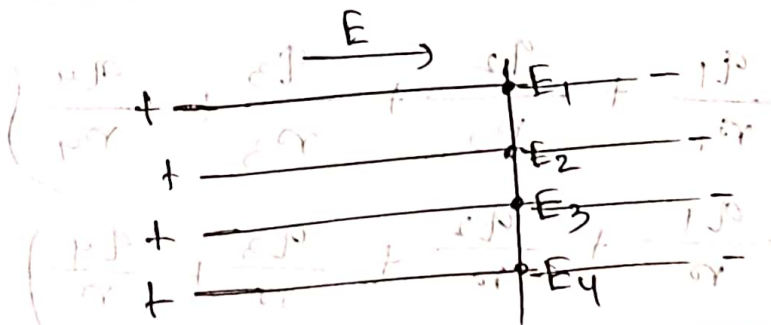
$$U = k \frac{q_1 q_2}{r}$$



$$U = \frac{k q q_0}{r_A} - \frac{k q q_0}{r_B}$$

Uniform Electric field:

- ① Lines are parallel to each other.
- ② Electric field strength is same.
- ③ Lines never cross each other.



$E_1 = E_2 = E_3$ in any cross section area then, we found uniform electric field.

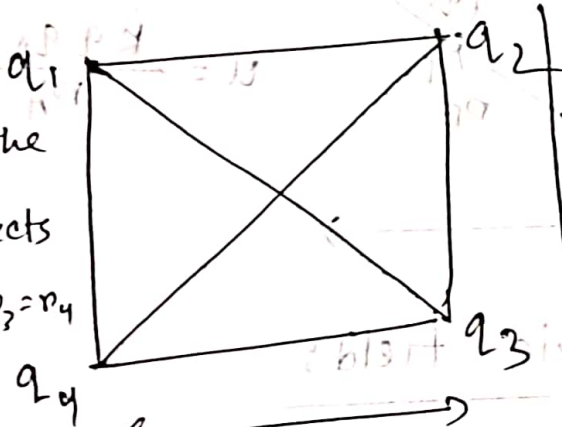
$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

$$V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r}$$

*** Ques: Show that for a point charge $V = k \frac{q}{r}$
 or for a group of point charge $V = k \sum_{i=1}^n \frac{q_i}{r_i}$

Problem 2:

For square where the two diagonals cross-sect there will be $r_1 = r_2 = r_3 = r_4$



Diagonal of square
 $r_1, r_3 = r = a\sqrt{2} = d\sqrt{2}$
 $= 1.3\sqrt{2}$

$$2r = 1.3\sqrt{2}$$

$$r = \frac{1.3\sqrt{2}}{2}$$

$$= \frac{1.3}{\sqrt{2}} \quad (1)$$

~~$V = k \sum \frac{q_i}{r_i}$~~

$$V = k \sum_{i=1}^4 \frac{q_i}{r_i}$$

$$= k \sum \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \frac{q_4}{r_4} \right)$$

$$= k \sum \left(\frac{q_1}{r} + \frac{q_2}{r} + \frac{q_3}{r} + \frac{q_4}{r} \right)$$

$$= k \frac{1}{r} \sum (q_1 + q_2 + q_3 + q_4)$$

$$= \frac{k}{r} (q_1 + q_2 + q_3 + q_4)$$

if the question is for electric force, electric field.

group of point charge:

$$\text{For Electric Field } E = \sum_{i=1}^N \frac{k q_i}{r_i^2}$$

$$= \frac{k}{r^2} (q_1 + q_2 + q_3 + q_4)$$

Electric Field has value & direction because it is a vector.

direction of Electric field:

$$E_1 + E_2 + E_3 + E_4 = E_{\text{net}}$$

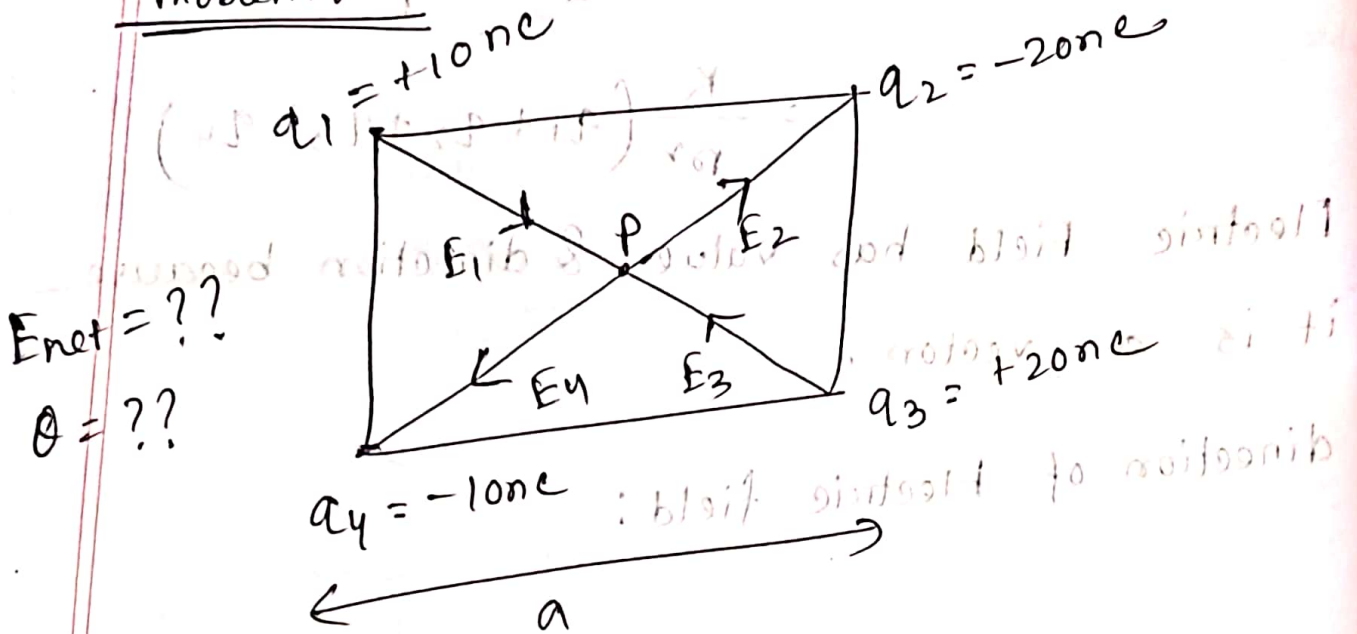
$$E_1 - E_2 = E_3$$

$$\frac{k q_1}{r_1^2} - \frac{k q_2}{r_2^2} = \frac{k q_3}{r_3^2}$$

if we want to find electric force for the same problem:

New Assignment

Problem: 11:



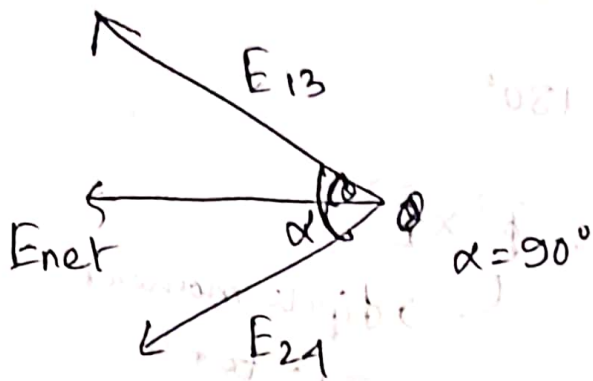
$$E_{net} = E_1 + E_2 + E_3 + E_4$$

$$E_{13} = E_3 - E_1$$

$$= \frac{kq_3}{r^2} - \frac{kq_1}{r^2}$$

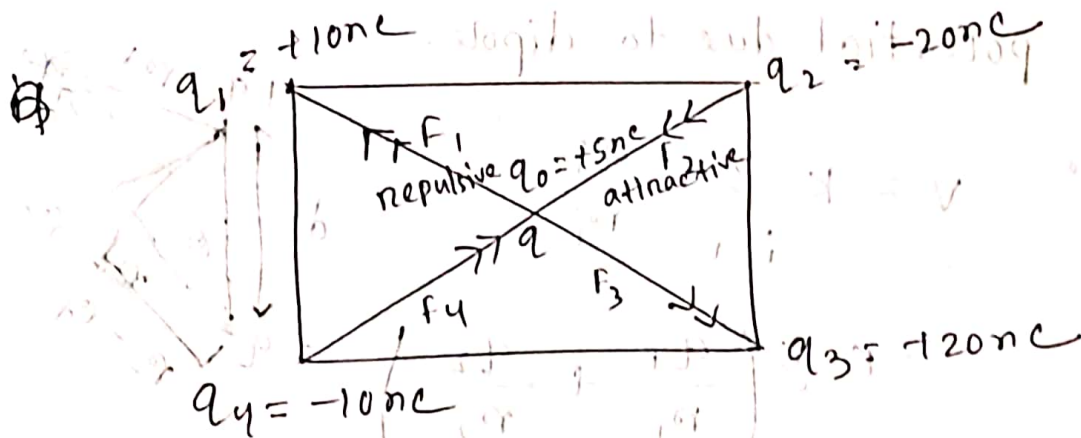
$$E_{24} = E_4 - E_2$$

$$= \frac{kq_4}{r^2} - \frac{kq_2}{r^2}$$



$$\theta = \tan^{-1} \frac{Q \sin \alpha}{P + Q \cos \alpha}$$

$$= \tan^{-1} \frac{E_{24} \sin 90^\circ}{E_{13} + E_{24} \cos 90^\circ}$$



$$q_0 = F_{\text{net}}$$

$$q_0 q_1 = F_1 = \text{outward} = \text{repulsive}$$

$$q_0 q_2 = F_2 = \text{inward} = \text{attractive}$$

$$q_0 q_3 = F_3 = \text{outward} = \text{repulsive}$$

$$q_0 q_4 = F_4 = \text{inward} = \text{attractive}$$

$$F_1 F_3 = q_1 q_3 = \theta = 180^\circ$$

$$F_2 F_4 = q_2 q_4 = \theta = 180^\circ$$

Ques: show that $\vec{\tau} = \vec{p} \times \vec{E}$
 &
 # $U = -\vec{p} \cdot \vec{E}$

dipole moment
 = (- to +)
 direction

$$V = \frac{p \cos \theta}{4\pi\epsilon_0 r}$$

Ques: show that $V = \frac{K p \cos \theta}{r^2}$

potential due to dipole.

Solve:

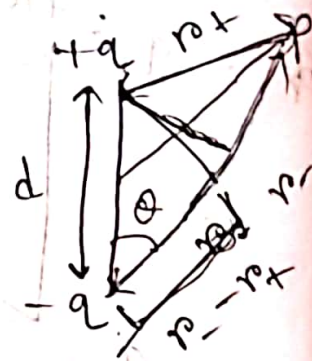
$$V = K \sum_{i=1}^2 \frac{q_i}{r_i}$$

$$= K \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} \right)$$

$$= K \left(\frac{+q}{r_+} + \frac{-q}{r_-} \right)$$

$$= Kq \left(\frac{1}{r_+} - \frac{1}{r_-} \right)$$

$$= Kq \left(\frac{(r_-) - (r_+)}{(r_+) (r_-)} \right)$$



$$= kq \times \frac{d \cos \theta}{(r_-) \times (r_+)}$$

$$(r_-) - (r_+) = d \cos \theta \quad \therefore = kq \times \frac{d \cos \theta}{r^2}$$

Ques: show that $V = Ed$ or $V = -Ed$

$$W = Fx = Fx \cos \theta$$

$$= \vec{F} \cdot \vec{x} = FS$$

$$dW = \vec{F} \cdot d\vec{s}$$

$$\int dW = \int q \vec{E} \cdot d\vec{s}$$

$$W = q \int \vec{E} \cdot d\vec{s}$$

$$= q \int_i^f \vec{E} \cdot d\vec{s} = q \int_i^f E \cdot ds$$

potential $\Rightarrow V = W/q$

$$W = q \int_i^f E \cdot ds \quad V = \int_i^f E \cdot ds$$

$$Vq = q \int_i^f E \cdot ds \quad V = \int E \cdot ds \cos \theta$$

$$V = Ed \quad = \int E ds \cos \theta = \int Ed$$

Assignment-3

12
10
45

2 equal $Q_1, 2$ equal q

$$\text{Potential, } V = \frac{1}{4\pi\epsilon_0} \times \frac{q}{r} = \frac{kq}{r}$$

$$\text{Potential energy, } U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r} = \frac{k q_1 q_2}{r}$$

$$\text{Electric field, } E = \frac{kq}{r^2}$$

$$E = \frac{F}{q_0}$$

$$\text{Potential} = \frac{1}{4\pi\epsilon_0} \frac{P \cos \theta}{r^2}$$

$$\tau = -PE \sin \theta$$

$$U = -PE \cos \theta$$

$$V_b = \int E \cdot d\vec{r}$$

$$p = qd$$

Electric Field

Q. Show that for the electric field due to electric dipole, $E = \frac{2kP}{z^3} = \frac{1}{4\pi\epsilon_0} \times \frac{2P}{z^3}$

Dipole in vertical arrangement:

$$r_+ = z - d/2$$

$$= z \left(1 - \frac{d}{2z} \right)$$

$$r_- = z + d/2$$

$$= z \left(1 + \frac{d}{2z} \right)$$

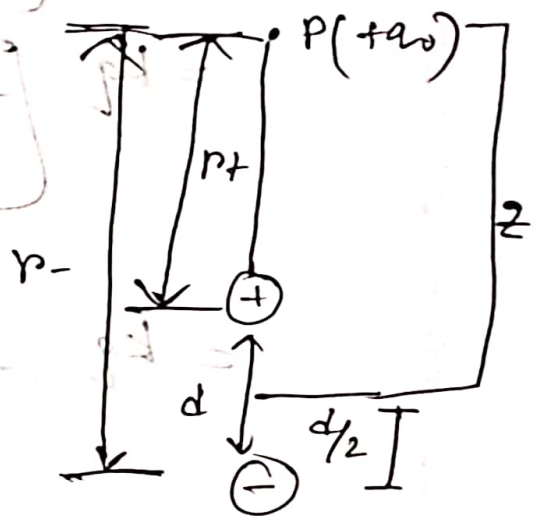
$$E_{\text{net}} = E_{(+)} + E_{(-)}$$

$$= E_{(+)} - E_{(-)}$$

$$= \frac{kq(+)}{r_+^2} - \frac{kq(-)}{r_-^2}$$

$$= \frac{kq}{r_+^2} - \frac{kq}{r_-^2}$$

$$= kq \left(\frac{1}{r_+^2} - \frac{1}{r_-^2} \right)$$



$$= kq \left[\frac{1}{\left\{ 2 \left(1 - \frac{d}{2z} \right) \right\}^2} - \frac{1}{\left\{ 2 \left(1 + \frac{d}{2z} \right) \right\}^2} \right]$$

$$= kq \left[\frac{1}{2^2 \left(1 - 2 \cdot \frac{d}{2z} + \frac{d^2}{4z^2} \right)} - \frac{1}{2^2 \left(1 + \frac{d}{z} + \frac{d^2}{4z^2} \right)} \right]$$

$$= kq \left[\frac{1}{2^2 \left(1 - \frac{d}{z} + \frac{d^2}{4z^2} \right)} - \frac{1}{2^2 \left(1 + \frac{d}{z} + \frac{d^2}{4z^2} \right)} \right]$$

$$= kq \frac{1 + \frac{d}{z} + \frac{d^2}{4z^2} - 1 + \frac{d}{z} - \frac{d^2}{4z^2}}{2^2 \left(1 - \frac{d}{z} + \frac{d^2}{4z^2} \right) \left(1 + \frac{d}{z} + \frac{d^2}{4z^2} \right)}$$

$$= kq \frac{2 \frac{d}{z}}{2^2 \left(1 - \frac{d}{z} + \frac{d^2}{4z^2} \right)}$$

$$= \frac{kq}{2^3} \frac{2d}{1 - \frac{d}{z} + \frac{d^2}{4z^2}}$$

$$= \frac{1}{2\pi\epsilon_0} \frac{q}{2^3} \frac{\left(1 - \left(\frac{d}{2z} \right)^2 \right)^{-2}}{1 \times d} \quad z \gg d$$

$z \gg d$ so that $\frac{d}{2z}$ will be neglected compared to 1.

$$E = \frac{q \times d}{2\pi\epsilon_0 2^3} = \frac{2Kqp}{2^3}$$

Q7) Find the electric field due to the ~~the~~ a line of a charge on

Show that $E = \frac{kq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$

Linear charge density / charge density

$$\lambda = \frac{\text{Charge}}{\text{length}}$$

$$\lambda = \frac{q}{L} \rightarrow S$$

$$\lambda = \frac{dq}{ds}$$

$$\lambda ds = dq$$

For Final Exam

Derivation

Assignment - 4: ~~is~~ Important. (3, 14, 15)

Question - 1: 4 set / 3 set short ques = $4 \times 2 = 8$

" - 2: ~~Math~~ 1 set Math = $8 \times 1 = 8$

" - 3: 1 set " = $8 \times 1 = 8$

" - 4: 1 set " = $8 \times 1 = 8$

" - 5: Broad question / prove / 1 set
show that = 2 ques (out of 2 set) ~~8~~ ~~10~~ ~~8~~
 $4 \times 4 = 8$

Derivation:

$$(i) E = \frac{2kP}{z^3}$$

$$(ii) E = \frac{kP}{z^2}$$

$$(iii) V = \frac{kP}{r} \quad V = k \sum_i \frac{q_i}{r_i}$$

$$(iv) V = \frac{kP \cos \theta}{r^2}$$

$$(v) V = Ed$$

$$(vi) \tau = \vec{E} \times \vec{P} \text{ and } u = -\vec{P} \cdot \vec{E}$$

(*) Uniform Electric field & characteristics.

(*) Equipotential surface.

(x, y, z)